

MemKernel: A Simple and Extensible Memory Service for Language Agents

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Abstract

Language agents require persistent memory to maintain user preferences, project decisions, and evolving facts across interactions. Storing complete conversation histories is simple, but it makes updates, retrieval, and provenance difficult to manage. We present **MemKernel**, a lightweight memory service that separates source events from durable memories. MemKernel uses a language model to extract atomic facts together with exact evidence spans, retrieves potentially related memories with vector similarity, and classifies each new fact as equivalent to, distinct from, or superseding an existing memory. The system stores active and historical memories in SQLite and retains links from every memory to its supporting source events. A REST API and an Agent Skill provide a small integration surface for language agents. In a preliminary retrieval-only evaluation on one LoCoMo conversation, MemKernel obtains an evidence hit rate of 83.33% and a mean evidence recall of 76.44% over 30 questions. These results are not an end-to-end question-answering evaluation, but they indicate that a compact and auditable memory pipeline can recover useful evidence from long conversations.

Keywords: agent memory, long-term memory, language agents, provenance, conversational retrieval

1 Introduction

Large language models operate on bounded input contexts, while practical assistants may interact with a user for weeks or months. Repeatedly supplying the full interaction history increases cost and introduces irrelevant context. Retrieval-augmented generation addresses part of this problem by combining a language model with an external index (Lewis et al., 2020). Long-term agent memory introduces additional requirements: the system must decide what is worth storing, update information that changes, and explain which interaction supports a remembered fact.

Existing work has explored hierarchical context management and learned memory pipelines. MemGPT treats context management as a hierarchy of memory tiers and moves information between them as needed (Packer et al., 2023). Mem0 extracts and consolidates conversational information before retrieval, emphasizing scalable deployment (Chhikara et al., 2025). Benchmarks such as LoCoMo and LongMemEval show that long-term memory requires more than keyword matching because questions may involve multiple sessions, temporal relations, knowledge updates, and abstention (Maharana et al., 2024; Wu et al., 2025).

MemKernel is designed as a small, inspectable memory component rather than a complete conversational agent. Its central design choice is to keep three objects separate: the original source event, the extracted durable fact, and the lifecycle state of that fact. This separation makes it possible to retrieve compact memories while preserving evidence and update history.

This paper makes three practical contributions:

1. It presents an evidence-bound extraction pipeline in which every generated memory must contain an exact supporting span from the current source event.
2. It introduces lifecycle-aware reconciliation with explicit EQUIVALENT, DISTINCT, and SUPERSEDES relations.
3. It provides a lightweight implementation based on FastAPI, SQLite, vector embeddings, and an Agent Skill integration layer.

2 Related Work

2.1 Retrieval and External Memory

Retrieval-augmented generation uses a non-parametric index to provide information that is not reliably accessible from model parameters alone (Lewis et al., 2020). This design supports inspection and updates, but a generic document index does not define which conversational facts should remain active when a user corrects or changes previous information.

MemGPT frames long contexts as a virtual memory problem and manages information across memory tiers (Packer et al., 2023). Mem0 instead builds a memory-centric pipeline that extracts, consolidates, and retrieves salient conversational information (Chhikara et al., 2025). MemKernel follows the extract-and-retrieve direction, but emphasizes a minimal service boundary, explicit memory states, and source-level provenance.

2.2 Long-Term Memory Evaluation

LoCoMo contains very long conversations spanning many sessions and evaluates question answering, summarization, and multimodal generation (Maharana et al., 2024). LongMemEval evaluates information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention (Wu et al., 2025). These benchmarks motivate memory systems that preserve temporal and update information rather than treating all past messages as equally current. The present work uses a small LoCoMo subset only as a preliminary retrieval check.

3 MemKernel

3.1 System Overview

MemKernel exposes two primary operations: *remember*, which processes a new source event, and *recall*, which retrieves relevant active or historical memories. Figure 1 summarizes the write and retrieval paths, including the shared provenance store. The processing path contains four stages.

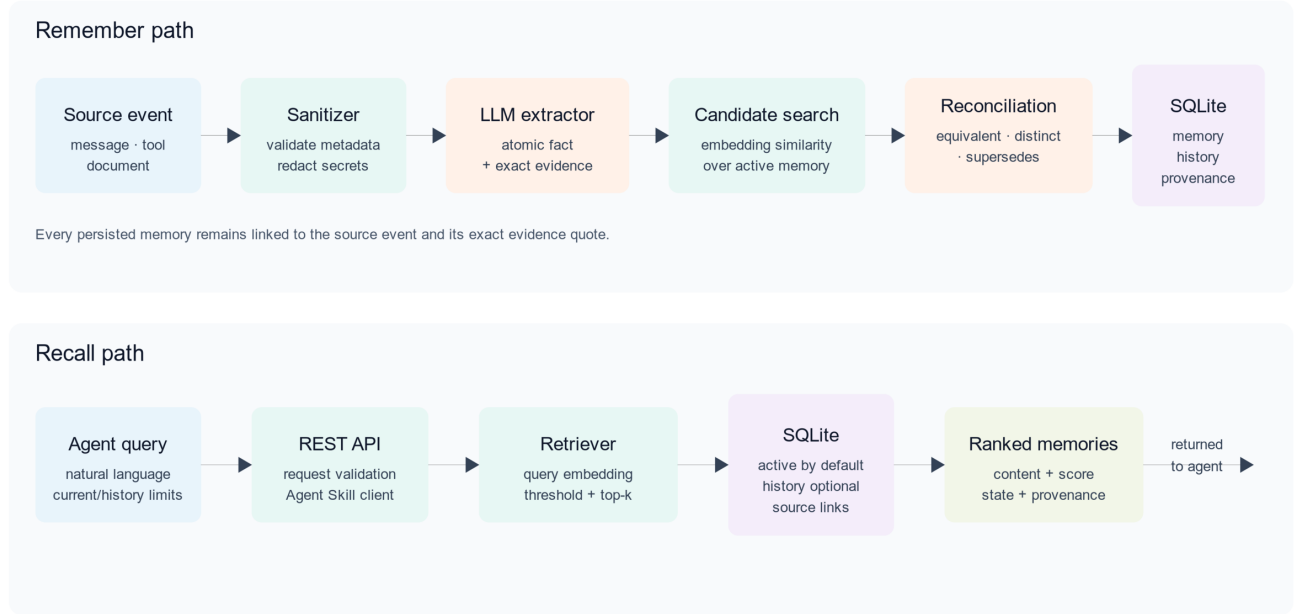


Figure 1. *MemKernel* system architecture. The remember path extracts and reconciles durable facts before an atomic write. The recall path searches active memories by default and can optionally return history and provenance.

Stage	Input	Main operation	Output
Source processing	Message, tool output, or document	Validate metadata and redact common secret patterns	Sanitized source event
Fact extraction	Source event and optional recent context	Extract atomic facts and exact evidence spans	Candidate memories
Reconciliation	Candidate fact and similar active memories	Classify relation and apply a lifecycle decision	Added, confirmed, or superseding memory
Retrieval	Natural-language query	Vector search and score filtering	Current and optional historical memories

The implementation separates model providers from storage and retrieval interfaces. Language-model and embedding backends can therefore be replaced without changing the public memory API.

3.2 Evidence-Bound Extraction

A source event contains an identifier, content, source type, role, observation time, and JSON metadata. Supported source types are messages, tool outputs, and documents. Before extraction, MemKernel applies best-effort redaction to common private keys, bearer tokens, API keys, and named secret fields.

The extractor asks a language model to return a small list of atomic and durable facts. Each fact contains content and evidence. The evidence must be a non-empty exact substring of the sanitized source content; otherwise, the result is rejected. This validation does not prove that the extracted fact is a perfect interpretation, but it prevents the system from storing a fact with no textual support in the current event.

The extractor may receive up to six recent messages through metadata. This context is used only to resolve pronouns, ellipsis, confirmations, and corrections. A fact still needs evidence in the current source event. For example, if the previous message asks whether SQLite should be used and the current message says "Yes, use it," recent context can resolve the referent, while the confirmation remains the evidence.

3.3 Reconciliation and Memory Lifecycle

Extracted facts are compared with a small set of active memory candidates selected by embedding similarity. A language model assigns one of three relations:

- **EQUIVALENT**: the new fact expresses the same durable claim;
- **SUPERSEDES**: the new fact concerns the same mutable claim and makes the previous value outdated;
- **DISTINCT**: both facts can remain active.

These relations map to three storage actions, as shown in Figure 2. A distinct fact creates a new active memory. An equivalent fact produces no duplicate memory and links the new source as confirmation. A superseding fact creates a new active memory and marks the old record as superseded. The backend applies the source event, memory updates, embeddings, and provenance links in one SQLite transaction.

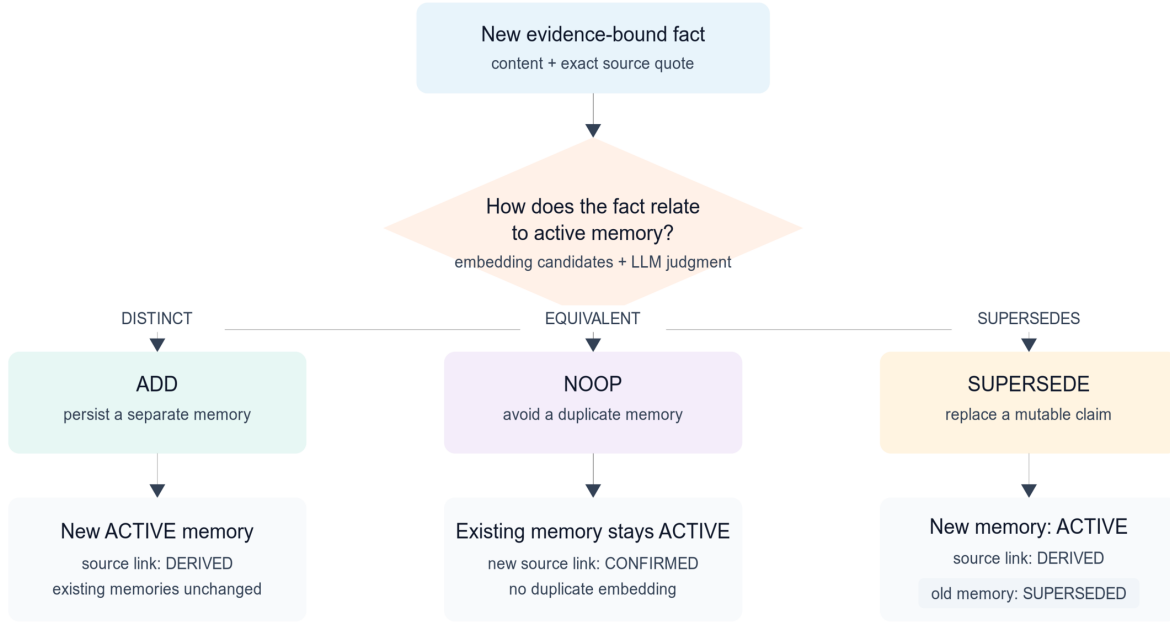


Figure 2. Reconciliation and memory lifecycle. Each extracted fact produces an explicit storage action while source links preserve whether the event derived or confirmed a memory.

Each memory record stores an identifier, content, creation time, state, and an optional pointer to its replacement. This representation retains a version chain without mixing outdated facts into default retrieval. Historical search remains available when an agent explicitly needs earlier values or the reason a value changed.

3.4 Provenance and Agent Integration

MemKernel stores source events separately from memories and connects them through a many-to-many provenance table. Each link includes the evidence quote and a link type: **DERIVED** for a source that created or replaced a memory, and **CONFIRMED** for a later source that matched an existing memory. An agent or developer can inspect both the complete memory history and the events supporting an individual memory.

The FastAPI service provides endpoints for remembering events, recalling memories, reading a memory’s history, and listing its sources. A bundled Agent Skill wraps these endpoints in a command-line client. Agents can therefore use MemKernel without embedding storage-specific logic in their prompts or application code.

4 Preliminary Evaluation

4.1 Setup

We ran a retrieval-only experiment on one conversation from LoCoMo (Maharana et al., 2024). The run ingested 369 dialogue turns as 19 session-bounded chunks and evaluated 30 questions. Retrieval returned up to 20 active memories per question with the similarity threshold set to zero. Answer generation and LLM-based judging were disabled, so the experiment measures evidence retrieval rather than question-answering accuracy.

For each question, the benchmark maps retrieved memories back to their linked LoCoMo dialogue identifiers. **Evidence hit rate** is the proportion of questions for which at least one gold evidence identifier is recovered. **Evidence recall** is the fraction of gold

evidence identifiers recovered for a question, averaged across questions with annotated evidence.

4.2 Results

Metric	Result
Questions	30
Evidence hit rate	83.33%
Mean evidence recall	76.44%
Average retrieval latency	32.79 ms

The result shows that the retrieved memories frequently preserve links to relevant source turns. Provenance is important for this measurement: the benchmark can evaluate evidence coverage without relying on lexical overlap between an extracted memory and a reference answer.

The numbers should be interpreted as a feasibility result. The run covers one conversation, uses a permissive zero similarity threshold, and does not compare against alternative memory systems. It also does not test whether a downstream language model can combine the retrieved memories into a correct answer.

5 Discussion

MemKernel’s main benefit is not a new retrieval algorithm, but a compact composition of extraction, updating, retrieval, and auditing. Evidence validation limits unsupported memory creation, explicit lifecycle states keep outdated information out of normal recall, and provenance links allow developers to inspect why a memory exists. The service boundary also makes the component usable by different agents and model providers.

The design trades breadth for simplicity. SQLite is appropriate for a local or single-node deployment but does not provide distributed scaling. Reconciliation depends on both embedding quality and a language-model judgment, so errors can merge distinct facts or preserve contradictory facts. Regex-based secret redaction is

best-effort and cannot replace application-level access control. The current API also requires deployment-specific authentication and authorization before it is exposed beyond a trusted local environment.

6 Limitations

The evaluation is preliminary and insufficient for claims of state-of-the-art performance. It includes only one LoCoMo conversation and 30 questions, omits full question answering, and reports no statistical uncertainty or baseline comparison. The retrieval latency is specific to the evaluated local configuration and should not be treated as a general performance guarantee. Future evaluation should cover all LoCoMo conversations, LongMemEval, stronger memory baselines, multiple model providers, end-to-end answer accuracy, and ablations of extraction context, reconciliation, and provenance-aware retrieval.

MemKernel currently represents durable information as natural-language facts. This improves readability but may be less suitable than structured graphs for relational or highly temporal queries. The three-way reconciliation label set also simplifies cases where information is partially compatible, uncertain, or valid only during a bounded time interval. As the interaction history grows, the memory database may expand without bound. MemKernel currently provides no principled deletion or compaction policy. Future work could investigate retention policies inspired by spaced-repetition forgetting schedules, together with hierarchical storage, to control storage growth and retrieval cost.

7 Conclusion

This paper introduced MemKernel, a small and extensible memory service for language agents. The system combines evidence-bound fact extraction, lifecycle-aware reconciliation, vector retrieval, and source provenance in a SQLite-backed service. A preliminary LoCoMo experiment suggests that this design can recover relevant evidence from a long conversation, while the limited evaluation leaves end-to-end effectiveness as future work. MemKernel provides a practical foundation for studying agent memory without requiring memory management to be tightly coupled to a specific agent framework.

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